

CHAPTER -1

INTRODUCTION

1.1 GENERAL INTRODUCTION

In today's competitive business environment, effective communication with customers plays a crucial role in achieving organizational goals. Telecalling remains one of the most widely used methods for customer interaction, lead generation, follow-ups, feedback collection, and sales conversion. Organizations across various domains such as education, banking, insurance, healthcare, marketing, and customer support rely heavily on telecalling teams to manage large volumes of customer data and maintain continuous engagement. However, managing telecalling operations efficiently requires systematic handling of lead data, status tracking, progress monitoring, and report generation.

With the increasing volume of customer data, traditional methods of telecalling management using manual registers or static spreadsheets have become inefficient and error-prone. These methods lack real-time visibility, structured lead tracking, and analytical insights. As a result, organizations face challenges such as duplicate data, inconsistent status updates, delayed reporting, and difficulty in monitoring telecaller performance. To overcome these limitations, there is a growing demand for intelligent and automated systems that can manage the complete lifecycle of telecalling leads in a structured and transparent manner.

1.2 BACKGROUND OF TELECALLING SYSTEMS

Telecalling systems have evolved significantly over the years. Initially, telecalling activities were managed manually, where call details and lead statuses were recorded on paper or basic spreadsheets. While this approach was suitable for small-scale operations, it became ineffective as organizations expanded and

the volume of leads increased. Manual tracking not only consumes time but also increases the risk of human errors, data loss, and miscommunication.

Modern telecalling operations require real-time access to lead data, quick status updates after each call, and centralized storage for better accessibility. Additionally, decision-makers require analytical insights such as the number of interested leads, follow-ups pending, conversions achieved, and unsuccessful calls. Without automation, generating such insights becomes a tedious and time-consuming process. This has led to the adoption of digital telecalling management systems that integrate data handling, visualization, and reporting functionalities.

1.3 PROBLEM STATEMENT

Despite the availability of basic digital tools, many organizations still face challenges in effectively managing telecalling leads. The commonly observed problems include lack of centralized data management, difficulty in tracking the lifecycle of leads, absence of visual representation of lead flow, and limited reporting capabilities. Telecallers often update lead statuses in separate files, making it difficult to maintain consistency and track real-time progress.

Moreover, management teams struggle to analyze telecalling performance due to the absence of status-based filtering and consolidated reporting. Downloading and sharing relevant data for analysis requires manual effort, which affects productivity and decision-making. These challenges highlight the need for a system that can automate telecalling workflows, provide real-time status visualization, and generate downloadable reports based on lead status.

1.4 NEED FOR THE PROPOSED SYSTEM

The need for an automated telecalling lead lifecycle tracking system arises from the growing complexity of telecalling operations. Organizations require a reliable platform that can handle large datasets, allow seamless status updates, and provide meaningful insights through visualization. A centralized system

reduces redundancy, ensures data integrity, and improves coordination between telecallers and administrators.

The proposed system addresses these needs by allowing users to upload telecalling data in spreadsheet format, view and update lead statuses dynamically, and track lead progress visually. The inclusion of flowchart-based visualization helps users understand the movement of leads across different stages. Additionally, the report download facility enables users to export filtered data or combine multiple statuses into a single sheet, making analysis and reporting more efficient.

1.5 OBJECTIVES OF THE PROJECT

The primary objectives of the proposed telecalling lead lifecycle tracking system are as follows:

- To design a centralized platform for managing telecalling lead data
- To enable easy upload and viewing of telecalling data using spreadsheets
- To provide real-time lead status updates
- To visualize lead progression using status flow representations
- To generate and download status-based reports
- To improve productivity and reduce manual effort in telecalling operations

1.6 SCOPE OF THE PROJECT

The scope of this project includes the design and development of a web-based telecalling management system that supports user authentication, data upload, status tracking, visualization, and report generation. The system is intended for small to medium-scale organizations that rely on telecalling for customer engagement.

The project focuses on managing telecalling leads efficiently and does not include advanced features such as predictive analytics or AI-based call

recommendations. However, the system is designed in a scalable manner, allowing future enhancements such as performance analytics, role-based access, and integration with CRM platforms.

1.7 PROPOSED SOLUTION OVERVIEW

The proposed solution is an end-to-end web-based telecalling lead lifecycle management platform designed to simplify, organize, and automate the entire telecalling workflow. The system provides a secure login mechanism to ensure that only authorized users can access telecalling data. Once logged in, users can upload telecalling lead data in Excel format, which is commonly used in organizations for storing customer and lead information. Each uploaded file can be assigned a specific campaign name or heading, enabling better identification and management of multiple telecalling campaigns simultaneously.

After uploading the data, the system displays the telecalling leads in a well-structured and user-friendly tabular format. This interface allows telecallers to easily view lead details such as customer name, contact number, and other relevant information. Following each call, telecallers can update the status of the lead directly within the system. Status categories such as Interested, Follow-up, Converted, Not Interested, and Not Reachable help in clearly identifying the current stage of each lead in the telecalling process.

The system continuously tracks all status updates and maintains the complete lifecycle history of every lead. To enhance clarity and understanding, the platform represents lead movement across different stages using visual charts and flow diagrams. These visual representations provide quick insights into the overall progress of telecalling campaigns and help administrators identify bottlenecks, pending follow-ups, and successful conversions. Visual status flow tracking also supports better monitoring of telecaller performance and campaign effectiveness.

In addition to visualization, the system offers powerful filtering and reporting capabilities. Users can filter leads based on their current status and generate reports accordingly. The report download facility allows users to export filtered data into Excel sheets for offline analysis, sharing, or record-keeping purposes. Furthermore, the system supports the merging of multiple status-based datasets into a single consolidated sheet. This feature enables comprehensive analysis by combining related lead statuses, thereby supporting informed decision-making and efficient reporting.

1.8 APPLICATIONS OF THE SYSTEM

The telecalling lead lifecycle tracking system can be applied in various domains, including:

- Sales and marketing organizations
- Educational institutions for admissions counseling
- Customer support and service centers
- Banking and financial services
- Healthcare appointment management
- Survey and feedback collection agencies

CHAPTER-2

LITERATURE SURVEY

2.1 TITLE: Web-Based Telecalling Management System

AUTHOR: R. Kumar

YEAR: 2021

DESCRIPTION:

This research presents a web-based telecalling management system developed to digitize and organize telecalling operations. The system allows users to store customer lead details in a centralized database and update call outcomes manually after each interaction. It aims to reduce dependency on paper-based records and improve accessibility by enabling data access through a web browser. The system provides a simple tabular view of leads, which helps telecallers track individual customer information efficiently.

However, the system lacks advanced analytical and visualization features that are essential for monitoring overall campaign performance. It does not support graphical representation of lead status distribution or workflow-based progress tracking. Report generation is limited and does not offer dynamic filtering or combined data exports. Compared to this system, our project introduces enhanced visualization, automated status flow tracking, and flexible report download facilities, making it more suitable for practical telecalling environments.

2.2 TITLE: CRM-Based Lead Tracking for Sales Automation

AUTHOR: S. Patel

YEAR: 2020

DESCRIPTION:

This study focuses on the implementation of Customer Relationship

Management (CRM) systems to automate sales and telecalling processes. The system supports features such as lead assignment, follow-up scheduling, interaction history maintenance, and status updates. CRM platforms are designed to improve customer engagement and sales productivity by maintaining detailed customer records.

Despite their effectiveness, CRM systems are often complex and require paid licenses, regular maintenance, and user training. Many small and medium-scale organizations find these systems difficult to adopt due to cost and infrastructure requirements. Additionally, CRM platforms include multiple modules that may not be directly relevant to telecalling operations. In contrast, our project delivers a simplified and cost-effective telecalling-specific solution that focuses only on essential features such as Excel-based lead upload, status tracking, visualization, and reporting.

2.3 TITLE: Spreadsheet-Based Lead Management System

AUTHOR: M. Joseph

YEAR: 2019

DESCRIPTION:

This approach discusses the use of spreadsheet applications such as Microsoft Excel for managing telecalling leads. Telecallers manually record customer details and update lead status after each call. Spreadsheets are widely used due to their simplicity, flexibility, and availability across organizations. They allow users to sort and filter data based on predefined columns.

However, spreadsheet-based systems suffer from several limitations when used for large-scale telecalling operations. They lack real-time synchronization, secure access control, and automated status tracking. Manual data entry increases the chances of human errors and data inconsistency. Additionally, visualization of lead progression is not available. Our project overcomes these limitations by

providing a secure web-based platform with automated status updates, real-time visualization, and downloadable reports.

2.4 TITLE: Online Lead Monitoring System Using Web Technologies

AUTHOR: A. Verma

YEAR: 2022

DESCRIPTION:

This research proposes an online lead monitoring system developed using web technologies such as HTML, CSS, JavaScript, and backend scripting. The system allows users to upload customer lead data and monitor call status through a browser-based interface. It improves accessibility and enables centralized data management across multiple users.

Although the system improves data availability, it lacks advanced analytical features required for decision-making. Visualization tools such as charts and flow diagrams are not integrated, and reporting capabilities are limited to basic data views. Our project extends this approach by integrating graphical status representation, advanced filtering options, and Excel-based report downloads, thereby providing a more comprehensive telecalling management solution.

2.5 TITLE: Data Visualization Techniques for Business Intelligence

AUTHOR: L. Anderson

YEAR: 2018

DESCRIPTION:

This study emphasizes the role of data visualization techniques in business intelligence systems. It explains how graphical representations such as charts, graphs, and dashboards help users analyze large datasets and identify trends effectively. Visualization improves understanding by converting raw data into meaningful visual insights.

While this research focuses on general business intelligence applications, its concepts are highly applicable to telecalling systems. Our project applies these visualization techniques specifically to telecalling lead management by displaying lead status distribution using charts. This enables administrators to quickly assess campaign performance, pending leads, and conversion rates, thereby improving operational efficiency.

2.6 TITLE: Secure Web Application Design with Authentication

AUTHOR: K. Sharma

YEAR: 2021

DESCRIPTION:

This work discusses secure web application development practices with a focus on authentication and authorization mechanisms. It highlights the importance of password encryption, session management, and access control in protecting sensitive data. The study emphasizes preventing unauthorized access and ensuring data confidentiality.

Although the research provides theoretical security guidelines, it does not apply them to a specific application domain. Our project practically implements these security concepts by integrating secure login, registration, OTP verification, password hashing, and session-based authentication. This ensures that telecalling data is accessible only to authorized users.

CHAPTER – 3

SYSTEM REQUIREMENTS

3.1 INTRODUCTION

System requirements define the essential conditions, technical specifications, and operational needs required for the successful development, deployment, and functioning of a software system. They serve as a foundation for the entire development process by clearly outlining what the system is expected to accomplish and the environment in which it should operate. Properly defined requirements help developers, designers, and stakeholders maintain a shared understanding of the system objectives and expected outcomes.

In the context of the telecalling lead lifecycle tracking system, system requirements are extremely important to ensure that the platform functions efficiently, handles data securely, and delivers reliable performance under different conditions. These requirements guide the design decisions, architecture selection, and implementation strategies of the system.

This chapter provides a detailed description of both functional and non-functional requirements, along with hardware and software specifications necessary for development and deployment. Additionally, it outlines system constraints and user expectations, which play a vital role in building a scalable, user-friendly, and efficient system capable of handling real-world telecalling operations.

3.2 FUNCTIONAL REQUIREMENTS

Functional requirements define the core operations and functionalities that the system must perform to meet user needs. These requirements describe how the system behaves in response to user inputs and how different components

interact with each other to achieve desired outcomes. They form the backbone of the system by specifying the exact features and services it must provide.

These requirements ensure that the system supports all necessary business processes involved in telecalling lead management. They also help developers understand how users will interact with the system and what actions they should be able to perform. By clearly defining functional requirements, the system can be designed to operate effectively and deliver expected results without ambiguity.

User Authentication

The system must provide a secure and reliable user authentication mechanism to ensure that only authorized users can access the platform. It should support user registration, login, and logout functionalities with proper validation checks. During registration, users should be able to create accounts using their email addresses, and the system should verify the authenticity of the email through an OTP-based verification process.

Passwords must be stored securely using strong encryption or hashing techniques to prevent unauthorized access and data breaches. The authentication system should also protect against common security threats such as brute force attacks and unauthorized login attempts. This ensures that user data remains safe and confidential at all times.

Login and Session Management

The system should allow registered users to log in using valid credentials such as email and password. Once authenticated, a secure session should be created and maintained throughout the user's interaction with the system. Session management plays a crucial role in ensuring that users remain authenticated while navigating between different pages.

The system should also implement session timeout mechanisms and secure session storage to prevent misuse. Unauthorized users must not be allowed to

access restricted areas such as dashboards, data management pages, or reports. Proper session handling enhances system security and ensures a seamless user experience.

Forgot Password and OTP Verification

The system should provide a secure and user-friendly password recovery mechanism for users who forget their login credentials. This feature should use an OTP-based verification process, where a one-time password is sent to the registered email address of the user.

Once the OTP is verified successfully, the system should allow the user to reset their password and regain access to their account. This process must be secure, time-bound, and resistant to unauthorized attempts. It improves user convenience while maintaining strong security standards.

Excel File Upload

The system must support uploading telecalling data through Excel files in a structured format. Users should be able to upload files easily by selecting them from their local system and associating them with specific campaigns or categories for better organization.

Before accepting the file, the system should validate the format, structure, and content of the Excel file to ensure compatibility. This helps prevent errors during data processing and ensures that only valid data is stored in the system. Proper file handling improves data consistency and reliability.

Data Viewing and Management

The system should display uploaded data in a well-organized and structured tabular format, making it easy for users to read and understand. Each record should include detailed information such as customer details, contact information, and current lead status.

Users should be able to navigate through the data efficiently, even when dealing with large datasets. The system should also support sorting and pagination to improve performance and usability. Effective data management ensures smooth operation and enhances user productivity.

Status Update Functionality

The system must allow users to update the status of each lead dynamically based on the outcome of telecalling interactions. Status options such as Interested, Follow-up, Converted, and Not Interested should be clearly defined and easily selectable.

Updates should be reflected in real time to ensure that the system always displays the most recent information. This functionality helps in tracking the progress of leads accurately and supports better decision-making. It also ensures that all team members have access to up-to-date information.

Data Visualization

The system should provide visual representations of data through charts and graphs to help users analyze lead performance effectively. Visual tools such as pie charts, bar graphs, and line charts can be used to display the distribution of lead statuses.

These visualizations make it easier for users to identify trends, patterns, and insights without manually analyzing raw data. Data visualization enhances decision-making and provides a clear overview of telecalling activities and outcomes.

Filtering and Searching

The system should include efficient filtering and searching functionalities to help users quickly locate specific data. Users should be able to filter leads based on status, campaign, or other relevant criteria.

This feature reduces the time required to find important information and improves overall efficiency. Advanced search capabilities ensure that users can access the exact data they need without navigating through large datasets manually.

Report Generation and Download

The system must allow users to generate reports based on filtered or selected data. These reports should be downloadable in Excel format for further analysis or sharing purposes.

Users should also have the option to combine multiple datasets into a single comprehensive report. This functionality supports better data analysis and helps in making informed business decisions. Report generation is essential for tracking performance and maintaining records.

File Management

The system should provide file management features that allow users to view, access, and delete uploaded files. Each user should only be able to manage their own files to ensure data privacy and security.

The system should also maintain proper organization of files, making it easy to locate and manage them. Efficient file management improves system usability and ensures better control over stored data.

3.3 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the overall quality attributes of the system, including performance, security, usability, and reliability. These requirements do not describe specific functionalities but focus on how well the system performs under various conditions.

They ensure that the system delivers a high-quality user experience, maintains stability, and operates efficiently. Non-functional requirements are

critical for ensuring that the system meets industry standards and user expectations.

Performance Requirements

The system should provide fast and responsive performance during all operations, including file uploads, data processing, and report generation. It should be capable of handling multiple user requests simultaneously without significant delays.

Efficient performance ensures smooth user interaction and prevents system slowdowns. Optimization techniques should be used to improve speed and responsiveness.

Security Requirements

The system must implement strong security measures to protect user data and prevent unauthorized access. Password encryption, secure authentication, and session management should be strictly enforced.

Additional security measures such as input validation and protection against common attacks should also be implemented. This ensures that the system remains safe and trustworthy.

Usability Requirements

The system should have an intuitive and user-friendly interface that is easy to navigate. Users should be able to perform tasks without requiring advanced technical knowledge.

Clear design, proper labeling, and simple workflows enhance usability and improve user satisfaction. A well-designed interface ensures efficient system usage.

Reliability Requirements

The system should function consistently without frequent failures or interruptions. It should ensure data integrity and provide mechanisms for error handling and recovery.

Reliable systems build user trust and ensure uninterrupted operations. Backup and recovery mechanisms should also be included.

Scalability Requirements

The system should be designed to handle increasing amounts of data and users as the application grows. It should maintain performance even under higher loads.

Scalability ensures long-term usability and supports business expansion without requiring major system changes.

Maintainability Requirements

The system should be easy to maintain, update, and modify. The codebase should be well-structured, modular, and documented.

This allows developers to implement new features and fix issues efficiently. Maintainability reduces long-term development costs and effort.

3.4 HARDWARE REQUIREMENTS

The telecalling lead lifecycle tracking system is designed to be lightweight and does not require high-end computing hardware. It can run efficiently on standard systems used in offices or personal environments.

Minimum Hardware Configuration

To ensure smooth and basic operation of the system:

- **Processor:** Intel Core i3 or equivalent
- **RAM:** Minimum 4 GB

- **Storage:** At least 10 GB free disk space
- **Display:** 1024×768 resolution
- **Input Devices:** Keyboard and mouse
- **Network:** Stable internet connection

Recommended Hardware Configuration

For better performance and efficient handling of large datasets:

- **Processor:** Intel Core i5 or higher
- **RAM:** 8 GB or more
- **Storage:** SSD with at least 20 GB free space
- **Display:** Full HD (1920×1080)
- **Network:** High-speed broadband internet

Optional Deployment Setup

- Web server for hosting the application
- Cloud platform for scalable deployment and remote access
- External or cloud-based backup storage for data safety

3.5 SOFTWARE REQUIREMENTS

The system is developed using modern and widely used technologies to ensure flexibility, scalability, and ease of maintenance. Most of the tools and frameworks used are open-source.

3.5.1 Backend Programming Environment

- **Runtime Environment:** Node.js
- **Framework:** Express.js
- **Operating Systems:** Windows, Linux, macOS
- **IDE/Editor:** VS Code or any modern code editor

3.5.2 Frontend Technologies

- **HTML** – Structure of web pages
- **CSS** – Styling and layout design
- **JavaScript** – Client-side scripting and interactivity
- **Bootstrap** – Responsive UI design and components

3.5.3 Database and Server

- **Database:** MongoDB (NoSQL database for storing user and lead data)
- **Server Environment:** Localhost for development and cloud server for deployment

3.5.4 Required Libraries and Packages

- **Multer** – File upload handling
- **XLSX** – Reading and processing Excel files
- **Nodemailer** – Sending emails and OTP verification
- **Bcrypt** – Password hashing and encryption
- **Express-session / JWT (optional)** – Session management and authentication

3.5.5 Browser Requirements

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Any modern web browser with JavaScript enabled

3.6 System Constraints

System constraints are limitations that may affect system design and performance. The system relies on internet connectivity for full functionality, which may limit usage in offline environments.

Excel files must follow a predefined structure to ensure compatibility. Storage limitations may affect handling of very large datasets. Security effectiveness depends on proper implementation and configuration of authentication mechanisms.

3.7 User Requirements

User requirements define the expectations and needs of end-users. Users expect a simple and efficient login and registration process that allows quick access to the system.

They require an easy-to-use interface for uploading files and managing data. Quick status updates and real-time feedback are essential for efficient workflow. Users also expect clear data visualization and easy report generation options. Secure handling of data is a critical requirement to ensure trust and reliability.

CHAPTER – 4

EXISTING SYSTEM

4.1 INTRODUCTION

The existing system refers to the currently used methods, tools, and techniques adopted by organizations to manage telecalling operations and maintain lead-related data before implementing the proposed solution. Studying the existing system is an important step in system analysis, as it helps in understanding how the current workflow operates and what problems are faced by users in real-world scenarios.

In many organizations, telecalling processes are still handled using traditional or semi-digital approaches. These systems often lack automation, real-time tracking, and proper integration of data. As a result, handling large volumes of customer information becomes complex, time-consuming, and prone to errors. The absence of centralized data management further complicates operations, making it difficult for teams to collaborate effectively. Additionally, these systems do not provide analytical insights, which are essential for improving performance and decision-making.

This chapter provides a detailed study of different types of existing systems used in telecalling operations, including manual systems, spreadsheet-based systems, and CRM-based systems. It also highlights their limitations and challenges, which justify the need for a more advanced and efficient solution.

4.2 MANUAL TELECALLING SYSTEM

In the manual telecalling system, all activities are performed using traditional methods such as notebooks, registers, and paper-based documents. Telecallers manually record customer details, responses, call history, and follow-

up schedules. This system does not require any digital infrastructure, making it simple and easy to implement for very small organizations.

Advantages of Manual System

- Easy to use without technical knowledge
- No setup or software cost required
- Suitable for small-scale operations
- No dependency on computers or internet

Limitations of Manual System

- Difficult to manage large volumes of data
- Time-consuming data entry and retrieval process
- High risk of data loss, damage, or duplication
- No backup or recovery mechanism
- No real-time updates or tracking
- Lack of data sharing between team members

Another major drawback of manual systems is the inability to perform data analysis. Managers cannot easily track lead progress, monitor telecaller performance, or generate reports. This leads to poor decision-making and reduces overall efficiency.

4.3 SPREADSHEET-BASED SYSTEM

With the introduction of digital tools, many organizations have shifted to spreadsheet-based systems using applications like Excel or similar software. In this method, telecalling data is stored digitally in tabular format, and telecallers update lead status manually.

Advantages of Spreadsheet System

- Better organization compared to manual records
- Easy data entry and editing
- Supports sorting and filtering
- Widely available and easy to use

- Low cost and accessible

Limitations of Spreadsheet System

- Manual data entry increases human errors
- Difficult to manage multiple files and sheets
- No centralized database for all leads
- Limited real-time collaboration features
- Risk of file corruption or accidental deletion
- Limited security and access control

In addition, spreadsheet systems provide only basic visualization and reporting features. Creating charts and reports requires manual effort, which consumes time and reduces productivity. As data grows, performance issues may also occur, making it difficult to handle large datasets efficiently.

4.4 CRM-BASED SYSTEMS

Customer Relationship Management (CRM) systems are advanced software solutions designed to manage customer interactions, sales processes, and lead tracking. These systems provide a centralized platform where all customer-related data can be stored and accessed.

Advantages of CRM Systems

- Centralized data management
- Automated follow-up reminders
- Improved communication tracking
- Built-in reporting and analytics
- Better organization of customer data
- Supports team collaboration

Limitations of CRM Systems

- High cost and subscription fees
- Requires technical knowledge and training
- Complex interface with unnecessary features

- Limited customization in some systems
- Implementation and maintenance challenges

Although CRM systems provide advanced features, they are often too complex for small and medium-scale telecalling operations. Many features are not directly relevant to telecalling, which makes the system difficult to use. Additionally, the cost factor makes it less accessible for smaller organizations.

4.5 LIMITATIONS OF THE EXISTING SYSTEM

The existing systems used in telecalling operations have several drawbacks that affect their efficiency, reliability, and usability.

Major Limitations

- Lack of real-time data tracking and updates
- Absence of proper data visualization tools
- High dependency on manual data entry
- Data inconsistency due to human errors
- Limited or no security mechanisms
- Difficulty in generating reports

Another major issue is the lack of integration between different systems. Data is often stored in multiple locations, making it difficult to combine and analyze. This leads to incomplete insights and poor decision-making.

Additionally, existing systems do not support automation, which increases workload and reduces productivity. These limitations clearly indicate the need for a more advanced system.

4.6 CHALLENGES IN EXISTING SYSTEMS

Organizations using traditional or existing telecalling systems face several practical challenges in their day-to-day operations.

Key Challenges

- Managing large volumes of lead data

- Difficulty in tracking lead progress
- Inefficient communication among team members
- Lack of centralized data storage
- Limited visibility for managers

Another important challenge is performance monitoring. Managers find it difficult to track telecaller activities such as number of calls made, successful conversions, and pending follow-ups. This reduces transparency and affects productivity.

Data duplication and inconsistency are also common problems, especially when multiple users work on the same data without proper coordination. These challenges negatively impact overall system efficiency.

4.7 NEED FOR IMPROVEMENT

The limitations and challenges identified in existing systems clearly highlight the need for an improved and modern solution. Organizations require a system that is efficient, reliable, and capable of handling large-scale telecalling operations.

Requirements for Improvement

- Automation of data handling processes
- Real-time data updates and tracking
- Secure authentication and data protection
- Centralized data management system
- Easy data upload and management
- Advanced visualization and reporting

The new system should also be user-friendly, ensuring that users with minimal technical knowledge can operate it easily. Cost-effectiveness is another important factor, especially for small and medium-scale organizations.

An improved system will help increase productivity, reduce errors, and enhance decision-making capabilities.

4.8 SUMMARY

This chapter provided a detailed analysis of the existing telecalling systems, including manual methods, spreadsheet-based systems, and CRM platforms. Each system has its own advantages but also suffers from several limitations.

Key Observations

- Manual systems are simple but inefficient
- Spreadsheet systems improve organization but lack automation
- CRM systems are powerful but expensive and complex

Overall, the existing systems fail to provide essential features such as automation, real-time tracking, secure data management, and efficient reporting. These drawbacks create a strong need for a more advanced telecalling lead lifecycle tracking system.

The next chapter will focus on the proposed system, which aims to overcome these limitations and provide a more efficient and user-friendly solution.

CHAPTER – 5

PROPOSED SYSTEM

5.1 INTRODUCTION

The proposed system, titled Telecalling Lead Lifecycle Tracking System with Status Flow Visualization and Report Download Facility, is designed to overcome the limitations of traditional telecalling management approaches and provide a modern, efficient solution. The system introduces a centralized, web-based platform that enables users to manage telecalling operations in a structured and organized manner.

In today's competitive business environment, telecalling plays a crucial role in customer engagement, lead generation, and conversion processes. However, traditional methods such as manual recording and spreadsheet management are not capable of handling large volumes of data efficiently. These methods often result in data inconsistency, duplication, and delays in updating information.

The proposed system addresses these issues by incorporating automation, real-time data processing, and centralized storage. It allows users to upload telecalling data in Excel format, eliminating the need for manual data entry. The system ensures that all data is stored securely and can be accessed anytime from a single platform.

Another important aspect of the system is dynamic status tracking. Users can update the status of leads instantly after each call, ensuring that the system always reflects the latest information. This improves coordination among team members and enhances overall workflow efficiency.

The system also provides visual representations of data through charts and status flow diagrams. These visual tools help users quickly understand lead distribution and campaign performance. In addition, the system supports report

generation and download features, enabling users to analyze data effectively and make informed decisions.

Overall, the proposed system improves productivity, reduces manual workload, enhances accuracy, and provides better insights into telecalling operations.

5.2 OBJECTIVES OF THE PROPOSED SYSTEM

The main objective of the proposed system is to develop an efficient, reliable, and user-friendly platform for managing telecalling activities and lead data.

Primary Objectives

- To design a centralized system for storing and managing telecalling data
- To automate data handling processes and reduce manual work
- To improve accuracy and consistency of stored data
- To provide real-time updates on lead status
- To enhance decision-making through data visualization

One of the key objectives is to eliminate the dependency on multiple files and disconnected systems by providing a single platform where all data can be accessed and managed. This ensures better organization and avoids confusion.

Another objective is to enable real-time lead tracking. Telecallers should be able to update lead status immediately, allowing managers to monitor progress without delay. This improves transparency and accountability within the team.

The system also aims to provide graphical insights through charts and flow visualization. These visual elements help users identify trends, track performance, and understand lead conversion patterns.

Additionally, the system focuses on generating customizable reports that can be downloaded and shared. This helps organizations analyze data efficiently and make data-driven decisions.

5.3 NEED FOR THE PROPOSED SYSTEM

The need for the proposed system arises due to the inefficiencies and limitations of existing telecalling management systems. Traditional methods fail to meet the requirements of modern organizations that deal with large volumes of customer data.

Key Reasons for the Need

- Manual systems are slow and error-prone
- Spreadsheet systems lack automation and scalability
- CRM systems are complex and expensive
- Lack of real-time data tracking
- Absence of proper visualization tools

Organizations require a solution that can handle large datasets efficiently while maintaining data accuracy. Existing systems increase workload due to repetitive manual tasks, which reduces productivity.

Another major need is improved data analysis. Managers require clear insights into telecalling performance, including lead status distribution, conversion rates, and pending follow-ups. Without proper tools, it becomes difficult to interpret data.

The proposed system fulfills these requirements by offering automation, real-time updates, secure data handling, and advanced reporting features. It provides a structured approach to telecalling management, making it essential for modern businesses.

5.4 FEATURES OF THE PROPOSED SYSTEM

The proposed system includes several advanced features that improve usability, efficiency, and performance.

Key Features

1. User Authentication

- Secure login and registration system

- OTP-based email verification
- Password encryption for data security

2. Excel File Upload

- Upload telecalling data in Excel format
- Automatic validation of file structure
- Campaign-wise file organization

3. Data Viewing Interface

- Tabular display of all records
- Easy navigation and readability
- Support for large datasets

4. Status Update Functionality

- Dynamic updating of lead status
- Multiple status categories (Interested, Follow-up, etc.)
- Instant reflection of changes

5. Data Visualization

- Charts and graphs for lead distribution
- Visual representation of status flow
- Easy understanding of performance metrics

6. Filtering and Searching

- Filter data based on lead status
- Quick search functionality
- Improved data accessibility

7. Report Generation

- Download filtered data as Excel files
- Combine multiple datasets into one report
- Easy sharing and analysis

These features collectively provide a complete solution for managing telecalling operations efficiently.

5.5 ADVANTAGES OF THE PROPOSED SYSTEM

The proposed system offers several advantages compared to traditional and existing systems.

Major Advantages

- **Automation:** Reduces manual work and increases efficiency
- **Accuracy:** Minimizes human errors in data handling
- **Real-Time Updates:** Ensures up-to-date information
- **Centralized Data Management:** All data stored in one place
- **Improved Productivity:** Faster operations and better workflow

Another major advantage is the availability of visualization tools, which help users understand data quickly. Charts and graphs provide insights that are difficult to obtain from raw data.

The system also improves collaboration among team members by providing a shared platform with updated information. Managers can monitor performance easily and take necessary actions.

Security is also enhanced through authentication and encryption techniques, ensuring that sensitive data is protected.

5.6 WORKING OF THE PROPOSED SYSTEM

The system operates through a structured workflow that ensures efficient data management and processing.

Step-by-Step Working

1. User Authentication

- User registers and logs into the system
- Secure session is created

2. Dashboard Access

- User accesses the main dashboard
- Options for upload, view, and manage data are displayed

3. Excel File Upload

- User uploads telecalling data file
- System validates and stores data

4. Data Display

- Data is shown in tabular format
- User can view all records clearly

5. Status Update

- User updates lead status after each call
- Changes are saved in real time

6. Data Visualization

- System generates charts based on lead status
- Visual insights are displayed

7. Filtering

- User filters data based on status
- Specific records are displayed

8. Report Generation

- User downloads filtered or combined data
- Reports are generated in Excel format

This workflow ensures smooth operation and efficient handling of telecalling data.

5.7 TOOLS AND TECHNOLOGIES USED

The proposed system is built using modern technologies that ensure high performance, scalability, and reliability.

Frontend Technologies

- **HTML:** Structure of web pages
- **CSS:** Styling and layout
- **JavaScript:** Interactivity and dynamic behavior
- **Bootstrap:** Responsive design

Backend Technologies

- **Node.js:** Server-side runtime environment
- **Express.js:** Web framework for handling requests

Database

- **MongoDB:** NoSQL database for storing data
- Flexible and scalable for large datasets

Libraries and Packages

- **Multer:** File upload handling
- **XLSX:** Excel file processing
- **Nodemailer:** Email and OTP sending
- **Bcrypt:** Password encryption
- **Session/JWT:** Authentication handling

Development Tools

- **Visual Studio Code:** Code editor
- Debugging and testing tools

Deployment Environment

- Localhost for development
- Cloud server for deployment

These technologies work together to create a robust and efficient system capable of handling modern telecalling operations.

CHAPTER – 6

SYSTEM DESIGN

6.1 INTRODUCTION

System design is the process of defining the architecture, components, modules, interfaces, and data flow of a system to satisfy specified requirements. It acts as a blueprint for the development of the system and provides a clear understanding of how different parts of the system interact with each other. It serves as a bridge between system requirements and actual implementation, ensuring that all functionalities are properly planned before development begins. A well-designed system helps developers visualize the overall structure and workflow, reducing confusion during coding and testing phases. It also ensures that the system is organized in a logical manner, making it easier to understand and manage. By clearly defining how components are connected and how data moves within the system, system design minimizes errors and improves development efficiency. Additionally, it helps in identifying potential issues early in the development process, saving time and effort in later stages.

In the proposed Telecalling Lead Lifecycle Tracking System, system design plays a crucial role in ensuring that the system is scalable, efficient, and easy to maintain. Since the system deals with large volumes of telecalling data and multiple user interactions, proper design is essential to handle operations smoothly. The design focuses on organizing the system in such a way that it can handle increasing data and user load without affecting performance. It also ensures that the system remains responsive and reliable under different working conditions. By carefully planning each component, the system becomes more structured and easier to manage. This approach also simplifies debugging and future enhancements, allowing developers to update or modify the system without

major difficulties. Overall, system design helps in building a strong foundation for the application.

The system follows a modular and layered approach, where the frontend handles user interactions, the backend processes requests, and the database manages data storage. In this structure, each layer has a specific role and responsibility, which helps in maintaining clarity and organization. The frontend layer is responsible for displaying information and capturing user inputs through an interactive interface. The backend layer acts as the core of the system, where all business logic and processing take place. It handles requests from the frontend, processes them, and communicates with the database. The database layer stores all system data securely and ensures efficient data retrieval when needed. This separation of concerns improves maintainability, as changes in one layer do not directly affect others. It also enhances system performance and scalability by allowing independent optimization of each layer.

The design also incorporates data flow mechanisms that ensure smooth communication between different components. When a user performs an action, such as uploading data or updating lead status, the request flows through the system in a structured manner. The frontend sends the request to the backend, where it is processed and validated. The backend then interacts with the database to store or retrieve the required information. Finally, the processed data is sent back to the frontend for display. This continuous and well-defined flow of data ensures that the system operates efficiently and without delays. Proper system design helps reduce complexity by organizing operations in a clear and systematic way. It also ensures that the system functions reliably under different conditions, providing consistent performance and a better user experience.

6.2 SYSTEM FLOW

The system flow represents the step-by-step process through which data moves within the system, starting from user input and ending with the final output

generation. It describes how different components of the system interact with each other to perform specific operations in a structured and organized manner. In the Telecalling Lead Lifecycle Tracking System, the flow begins when the user logs into the system and accesses the dashboard. From there, the user can upload telecalling data, which is then processed by the backend and stored in the database. The system retrieves and displays this data in a tabular format for easy viewing and management. When the user updates the status of a lead, the changes are sent to the backend, updated in the database, and reflected in real time on the user interface. The system also processes this data to generate visual representations such as charts, providing insights into lead distribution and performance. Additionally, users can apply filters to extract specific data and generate downloadable reports. Throughout this process, data flows seamlessly between the frontend, backend, and database layers, ensuring efficient communication and consistent data handling. A well-defined system flow helps in reducing complexity, improving performance, and ensuring that all operations are executed accurately and efficiently within the system.

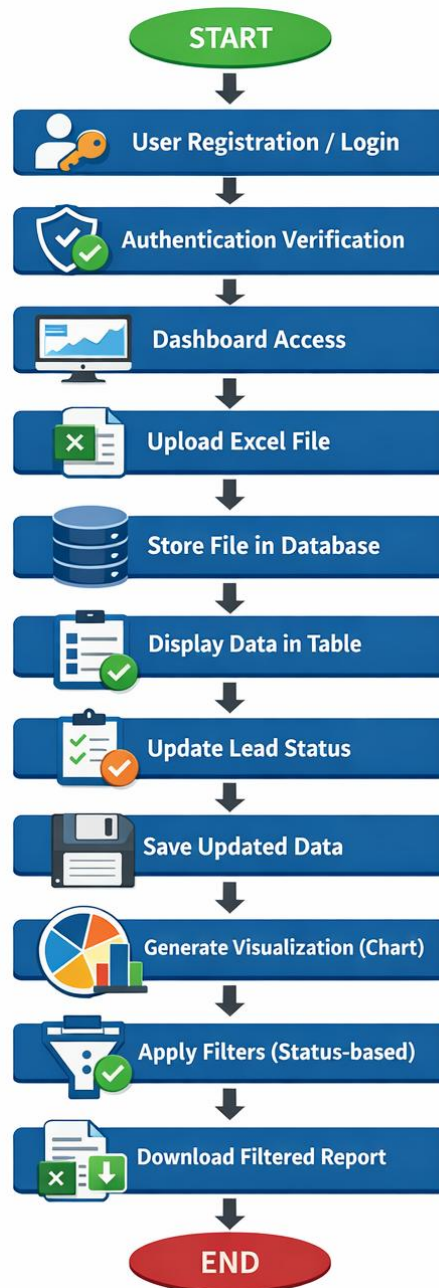


Fig 6.2 System Flow Chart

Step-by-Step Explanation

Step 1: User Registration/Login

The process begins when a user accesses the system and either registers as a new user or logs in as an existing user. During registration, the user provides necessary details such as email and password, which are securely stored in the system. For existing users, login is performed using valid credentials. This step ensures that only authorized users can access the system functionalities. It acts as

the entry point for all operations and helps in maintaining user-specific data and activities within the platform.

Step 2: Authentication Verification

Once the user submits login credentials, the system verifies them using secure authentication mechanisms. Passwords are compared with encrypted values stored in the database to ensure security. If the credentials are valid, the system grants access and creates a user session. If the credentials are invalid, access is denied, and an error message is displayed. This step ensures data protection and prevents unauthorized access to sensitive information within the system.

Step 3: Dashboard Access

After successful authentication, the user is redirected to the dashboard, which serves as the main interface of the system. The dashboard provides access to all major features such as file upload, data viewing, status updates, and report generation. It is designed to be user-friendly and intuitive, allowing users to navigate easily between different functionalities. The dashboard also provides a summary view of telecalling data and system activities.

Step 4: Upload Excel File

The user uploads a telecalling dataset in Excel format through the provided interface. The file typically contains customer details such as name, contact number, and lead information. The system checks the file format and structure to ensure compatibility before accepting it. This step eliminates the need for manual data entry and allows users to quickly import large volumes of data into the system.

Step 5: Store File in Database

After successful upload, the system stores the file on the server and saves relevant metadata such as file name, upload time, and user information in the database. This ensures proper organization and easy retrieval of files. The storage process is handled securely to prevent data loss or unauthorized access. This step enables efficient management of uploaded datasets.

Step 6: Display Data

The system reads the uploaded Excel file and extracts its contents for processing. The data is then displayed in a structured tabular format on the user interface. Each row represents a lead, and columns display details such as customer information and status. The table is designed to be clear and easy to navigate, allowing users to quickly view and analyze the data.

Step 7: Update Lead Status

Users can update the status of each lead based on the outcome of telecalling interactions. Status options such as Interested, Follow-up, Converted, and Not Interested are provided. This step allows real-time tracking of lead progress and ensures that the system always reflects the latest information. It also helps in organizing leads based on their current stage.

Step 8: Save Updated Data

Once the user updates the lead status, the system saves the changes in the database immediately. The updated information is reflected in real time on the interface, ensuring consistency across the system. This step ensures that no data is lost and that all updates are accurately recorded for future reference and analysis.

Step 9: Generate Visualization

The system processes the updated data to generate graphical representations such as pie charts or bar graphs. These visualizations display the distribution of lead statuses, helping users understand overall performance. Visualization makes it easier to identify trends, patterns, and areas that require attention, improving decision-making.

Step 10: Apply Filters

Users can apply filters to view specific categories of data based on lead status or other criteria. For example, users can filter only “Interested” or “Follow-up” leads. This functionality helps in quickly accessing relevant data without going through the entire dataset. It improves efficiency and saves time.

Step 11: Download Report

Finally, the system allows users to download filtered or combined data as an Excel report. These reports can be used for further analysis, sharing, or record-keeping. The report generation process is automated, ensuring accuracy and consistency. This feature supports better decision-making and helps in tracking telecalling performance effectively.

6.3 SYSTEM ARCHITECTURE

The system architecture defines how different components of the system are structured and how they interact with each other to deliver the required functionality in an efficient and organized manner. In the Telecalling Lead Lifecycle Tracking System, the architecture follows a layered approach consisting of the frontend, backend, and database layers. The frontend layer is responsible for handling user interactions and presenting data through a user-friendly interface, while the backend layer processes user requests, implements business logic, and manages communication between the frontend and the

database. The database layer stores all relevant information such as user details, uploaded files, and lead data in a secure and structured format. This architecture ensures smooth data flow, better performance, and easy scalability. It also improves maintainability by allowing independent development and modification of each layer without affecting the entire system. Overall, the system architecture provides a clear framework that supports efficient operation, secure data handling, and reliable system performance.

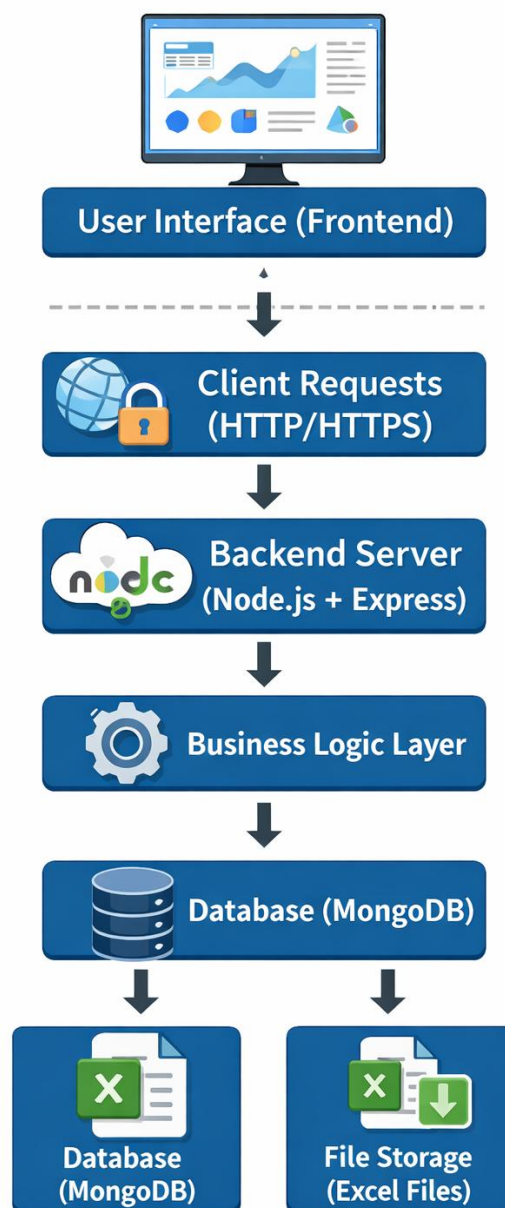


Fig 6.3 System Architecture

6.3.1 Architecture Explanation

The system follows a three-tier layered architecture, which divides the entire application into three distinct layers: Presentation Layer, Application Layer, and Data Layer. This structured approach ensures that each part of the system performs a specific function, improving organization, efficiency, and maintainability. By separating the system into layers, the complexity of development is reduced, and each layer can be developed and managed independently without affecting others.

1. Presentation Layer (Frontend)

The Presentation Layer is the topmost layer of the system and is responsible for handling all user interactions. It acts as the interface between the user and the system, allowing users to perform various operations such as logging in, uploading files, viewing data, updating lead status, and generating reports. This layer is designed using technologies such as HTML, CSS, JavaScript, and Bootstrap to create an interactive and user-friendly interface.

The frontend ensures that information is displayed clearly and in an organized manner through tables, forms, charts, and buttons. It captures user inputs and sends requests to the backend for processing. Additionally, it provides validation for user inputs to reduce errors before sending data to the server. The design of this layer focuses on usability, responsiveness, and accessibility so that users can operate the system easily on different devices. Overall, the presentation layer plays a key role in enhancing user experience and ensuring smooth interaction with the system.

2. Application Layer (Backend)

The Application Layer, also known as the backend, is the core part of the system where all business logic and processing take place. It is implemented using Node.js and Express.js, which handle server-side operations and manage

communication between the frontend and the database. This layer receives requests from the frontend, processes them, and sends appropriate responses back to the user interface.

The backend performs important operations such as user authentication, session management, file upload handling, data processing, status updates, and report generation. It ensures that all operations are executed correctly and securely. This layer also validates incoming data, applies business rules, and interacts with the database to store or retrieve information.

Another important function of the application layer is ensuring system security by implementing authentication and authorization mechanisms. It prevents unauthorized access and protects sensitive data. By handling all core functionalities, this layer ensures that the system operates efficiently and reliably.

3. Data Layer (Database)

The Data Layer is responsible for storing, managing, and retrieving all system data. In this system, MongoDB is used as the primary database to store user information, file metadata, and lead-related details. The actual Excel files uploaded by users are stored on the server, while their references and details are maintained in the database.

This layer ensures that data is stored in a structured and organized manner, allowing efficient access and retrieval when needed. It supports operations such as inserting new data, updating existing records, and deleting unwanted data. The database is designed to handle large volumes of data while maintaining performance and reliability.

Data security is also an important aspect of this layer. Proper measures such as access control and secure connections are implemented to protect data from unauthorized access. Overall, the data layer acts as the backbone of the system, ensuring data integrity, consistency, and availability.

6.3.2 Benefits of Layered Architecture

The layered architecture used in the system provides several advantages that improve the overall performance, flexibility, and maintainability of the application. By dividing the system into separate layers, each component can function independently, making the system more organized and efficient.

Key Benefits

1. Separation of Concerns

Each layer is responsible for a specific set of tasks, such as user interaction, business logic, or data storage. This separation makes the system easier to understand, develop, and manage. Developers can focus on one layer at a time without worrying about other parts of the system.

2. Scalability

The layered architecture allows the system to scale efficiently as the number of users or data increases. Individual layers can be upgraded or expanded without affecting the entire system. For example, the database can be optimized or the backend can be enhanced to handle higher loads.

3. Maintainability

Since the system is divided into separate layers, debugging and maintenance become easier. If an issue occurs, it can be identified and fixed within a specific layer without impacting the whole system. This reduces development time and effort.

4. Security

Sensitive operations such as authentication and data processing are handled in the backend layer, which is more secure than the frontend. This reduces the risk of unauthorized access and ensures better protection of user data.

5. Flexibility

The architecture allows easy modification and enhancement of the system. New features can be added, and existing functionalities can be updated without

major changes to the overall system structure. This makes the system adaptable to future requirements.

6. Reusability

Components developed in one layer can be reused in other parts of the system or even in different projects. This improves development efficiency and reduces redundancy.

7. Improved Performance

Each layer can be optimized independently, which helps in improving overall system performance. Efficient communication between layers ensures faster processing and better user experience.

Overall, the layered architecture provides a strong foundation for building a scalable, secure, and efficient system. It ensures that the Telecalling Lead Lifecycle Tracking System can handle real-world requirements while remaining flexible for future improvements.

6.4 MODULES

The system is divided into multiple modules, where each module is designed to handle a specific functionality of the application. This modular approach improves system organization, simplifies development, and enhances maintainability. Each module works independently but interacts with other modules to ensure smooth system operation. By separating functionalities into modules, the system becomes more scalable, flexible, and easier to debug or upgrade. This structure also allows developers to modify or enhance individual modules without affecting the entire system.

6.4.1 Authentication Module

The Authentication Module is responsible for managing user access and ensuring that only authorized users can use the system. It handles key operations such as user registration, login, logout, and password recovery. During

registration, users provide necessary details such as email and password, which are securely stored in the database using encryption techniques like hashing.

This module also implements OTP-based verification to confirm the authenticity of user email addresses. During login, the system validates user credentials and creates a secure session to maintain user access. In case users forget their passwords, the module provides a password recovery feature using OTP verification, ensuring a secure reset process.

Additionally, this module plays a crucial role in protecting sensitive data by preventing unauthorized access. It ensures proper authentication and authorization mechanisms, making the system secure and reliable. Overall, the authentication module forms the foundation of system security and user management.

6.4.2 File Upload Module

The File Upload Module allows users to upload telecalling data in Excel format into the system. It provides a user-friendly interface where users can select and upload files easily. Before accepting the file, the module performs validation checks to ensure that the file format and structure meet the required specifications.

Once validated, the file is securely stored on the server, and its metadata, such as file name, upload date, and associated user, is stored in the database. This helps in organizing and managing uploaded files efficiently.

This module eliminates the need for manual data entry by allowing bulk data upload, saving time and effort. It also ensures data consistency by accepting only properly structured files. Proper error handling is implemented to notify users in case of invalid files or upload failures. Overall, this module plays a key role in efficient data input and management.

6.4.3 Data Processing Module

The Data Processing Module is responsible for reading and processing the uploaded Excel files. It uses specialized libraries to extract data from Excel sheets and convert it into a structured format suitable for display and further processing.

This module ensures that data is cleaned, validated, and organized before being used by other modules. It handles tasks such as parsing rows and columns, mapping data fields, and converting raw data into usable formats. The processed data is then stored or passed to the frontend for display in a tabular format.

By automating data processing, this module reduces manual effort and minimizes errors. It ensures that large datasets can be handled efficiently without performance issues. Additionally, it prepares data for further operations such as status updates, filtering, and report generation. This module is essential for maintaining data accuracy and consistency throughout the system.

6.4.4 Status Management Module

The Status Management Module is responsible for updating and managing the status of telecalling leads. It allows users to modify the status of each lead based on the outcome of telecalling interactions. Status categories such as Interested, Follow-up, Converted, and Not Interested are supported.

This module ensures that all updates are processed in real time and reflected immediately in the system. It maintains data consistency by ensuring that changes are accurately stored in the database. Users can easily track the progress of leads and identify their current stage in the lifecycle.

The module also helps in organizing data by categorizing leads based on their status. This improves data clarity and supports better decision-making. By providing dynamic status updates, this module enhances workflow efficiency and ensures that all information remains up-to-date.

6.4.5 Visualization Module

The Visualization Module is responsible for presenting data in graphical formats such as charts and graphs. It processes the lead data and generates visual representations like pie charts and bar graphs to show the distribution of lead statuses.

This module helps users quickly understand complex data without analyzing raw tables. Visual representation makes it easier to identify trends, patterns, and performance metrics. For example, users can easily see how many leads are converted, pending, or not interested.

The module updates visualizations dynamically whenever data changes, ensuring that users always see the latest information. This improves decision-making and helps managers evaluate telecalling performance effectively. Overall, the visualization module enhances data analysis and user experience.

6.4.6 Filtering and Reporting Module

The Filtering and Reporting Module provides features for extracting specific data and generating reports. Users can apply filters based on lead status or other criteria to view only relevant records. This helps in quickly identifying important data without going through the entire dataset.

The module also allows users to generate reports based on filtered data. These reports can be downloaded in Excel format for further analysis or sharing. Additionally, users can combine multiple filtered datasets into a single report for comprehensive insights.

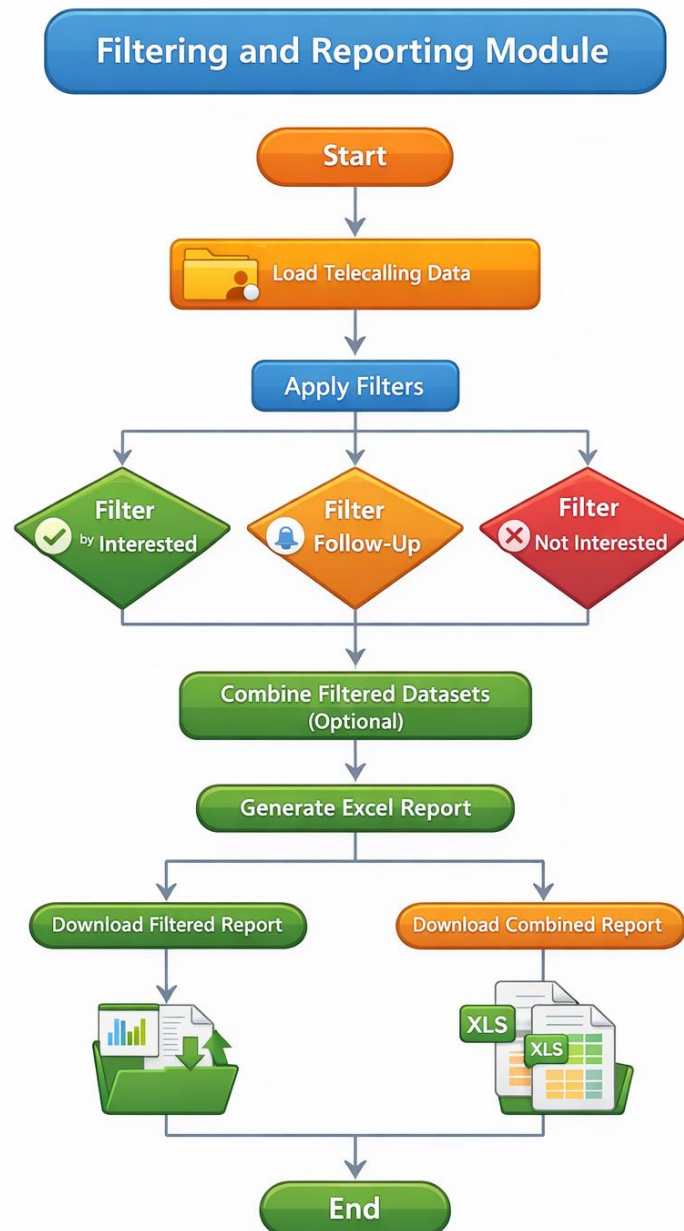


Fig 6.4.6 Filtering and Reporting Module

This module improves efficiency by automating report generation and reducing manual effort. It supports better decision-making by providing structured and organized data. The ability to filter and download reports makes this module highly useful for analysis and record-keeping.

6.4.7 File Management Module

The File Management Module is responsible for handling all operations related to uploaded files. It allows users to view, access, and manage their

uploaded files within the system. Users can see a list of all uploaded files along with details such as file name and upload date.

This module also provides the ability to delete files that are no longer needed. It ensures that users can only access and manage their own files, maintaining data privacy and security. Proper file organization helps users easily locate and manage their data.

Additionally, this module helps in maintaining system storage by allowing users to remove unnecessary files. It ensures efficient file handling and improves overall system usability. By providing complete control over uploaded data, this module enhances user experience and system management.

6.5 Algorithms and Techniques Used

The system uses several algorithms and techniques to perform its core operations efficiently and securely. Each algorithm is designed to handle a specific functionality such as authentication, file processing, data filtering, and report generation. These algorithms ensure that the system operates smoothly, reduces manual effort, and improves overall performance. By implementing structured algorithms, the system achieves better accuracy, reliability, and scalability.

6.5.1 Authentication Algorithm

Input: Email and Password

Process:

- Fetch user from database
- Compare hashed password
- If match → Login success

Output: Access granted/denied

The Authentication Algorithm is responsible for verifying user identity and ensuring secure access to the system. It works by retrieving user details from the

database and comparing the entered password with the stored hashed password using encryption techniques. This ensures that passwords are never stored in plain text, enhancing security.

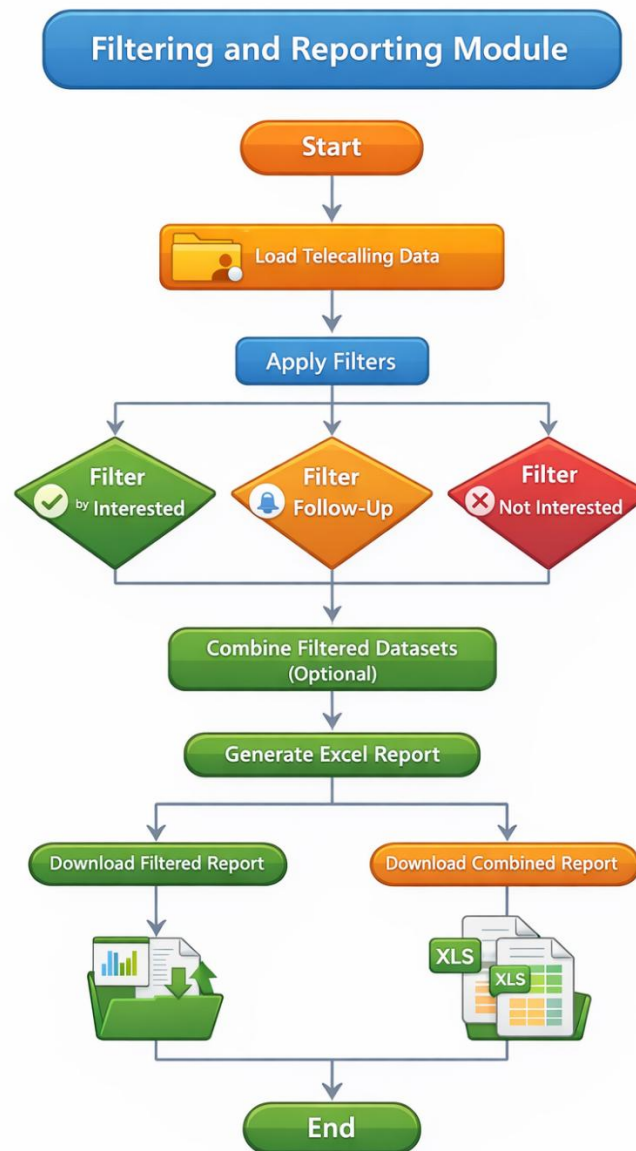


Fig 6.5.1 Authentication Algorithm

This algorithm prevents unauthorized access and protects sensitive user data. It also supports secure session management, allowing users to remain logged in during their interaction with the system.

Advantages

- Ensures high-level security through password encryption
- Prevents unauthorized system access

- Protects user credentials from data breaches
- Provides reliable and secure login mechanism

6.5.2 OTP Verification Algorithm

Process:

- Generate random 6-digit OTP
- Store OTP temporarily
- Send via email
- Compare user input with stored OTP

The OTP Verification Algorithm adds an additional layer of security during user registration and password recovery. It generates a unique one-time password and sends it to the user's registered email address. The system then verifies the OTP entered by the user within a limited time frame.

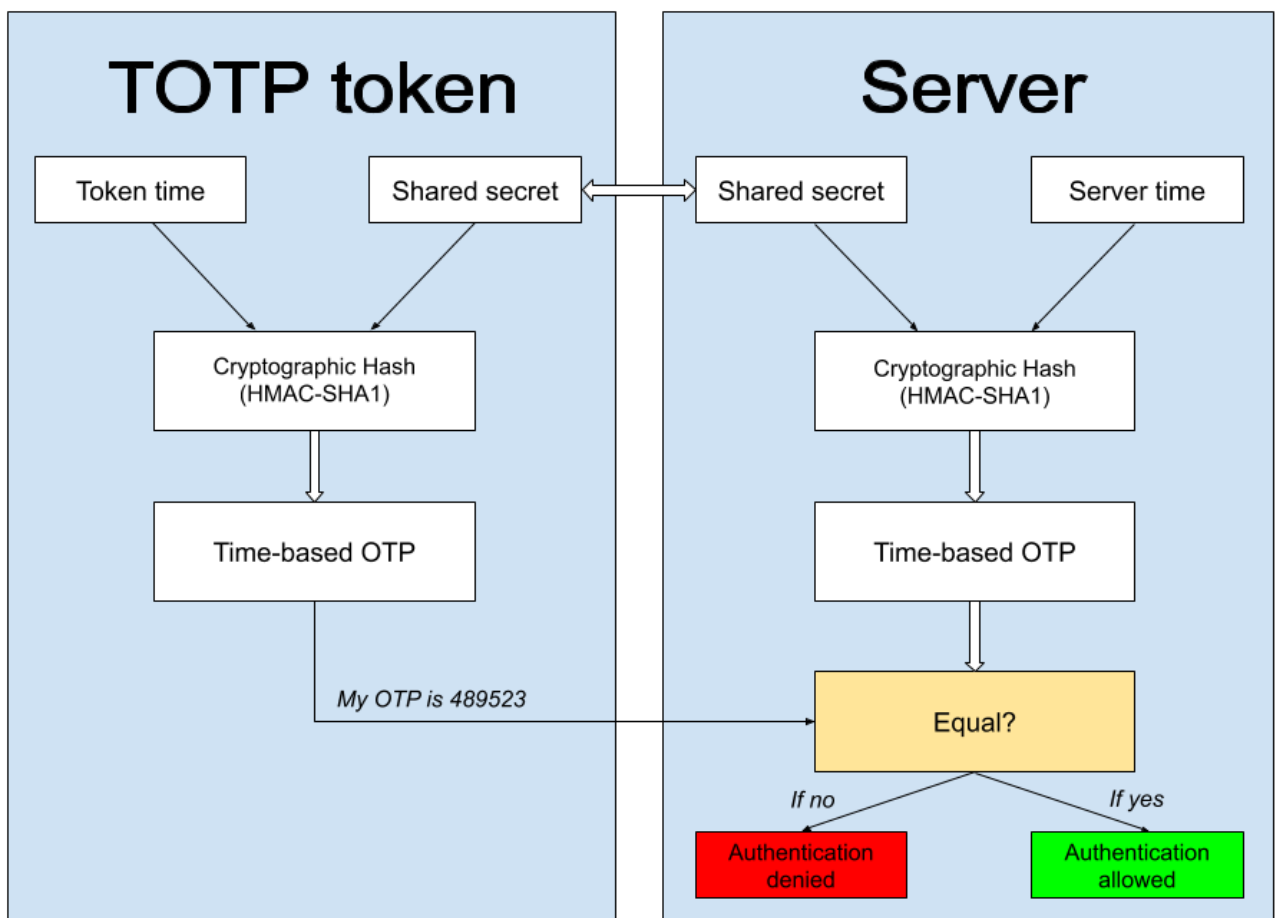


Fig 6.5.2 OTP Verification Algorithm

This algorithm ensures that only authorized users can complete sensitive actions such as account creation or password reset. It enhances system security by preventing unauthorized access even if login credentials are compromised.

Advantages

- Provides two-factor authentication
- Enhances account security
- Prevents unauthorized password reset
- Ensures user identity verification

6.5.3 File Upload Algorithm

Process:

- Accept file input
- Validate format (.xlsx/.xls)
- Store file in server
- Save metadata in database

The File Upload Algorithm handles the process of uploading telecalling data into the system. It ensures that only valid Excel files are accepted by checking the file format and structure. Once validated, the file is securely stored on the server, and its metadata is recorded in the database.

This algorithm simplifies data input by allowing bulk upload of records, eliminating the need for manual data entry. It also ensures that the uploaded data is properly organized and easily retrievable.

Advantages

- Supports bulk data upload
- Reduces manual data entry effort
- Ensures data consistency and accuracy
- Provides secure file storage mechanism

6.5.4 Status Update Algorithm

Input: Row index + new status

Process:

- Update corresponding record
- Save updated data
- Reload data for display

The Status Update Algorithm allows users to modify the status of telecalling leads dynamically. It identifies the specific record using the row index and updates the status based on user input. The updated data is then saved in the database and reflected immediately on the user interface.

This algorithm ensures real-time updates, enabling users to track lead progress efficiently. It maintains data consistency and ensures that all users have access to the latest information.

Advantages

- Enables real-time data updates
- Improves tracking of lead lifecycle
- Maintains data accuracy and consistency
- Enhances workflow efficiency

6.5.5 Data Filtering Algorithm

Input: Selected statuses

Process:

- Loop through dataset
- Match status condition
- Store filtered results

The Data Filtering Algorithm is used to extract specific records from the dataset based on selected criteria such as lead status. It iterates through the dataset, checks for matching conditions, and stores the filtered results for display.

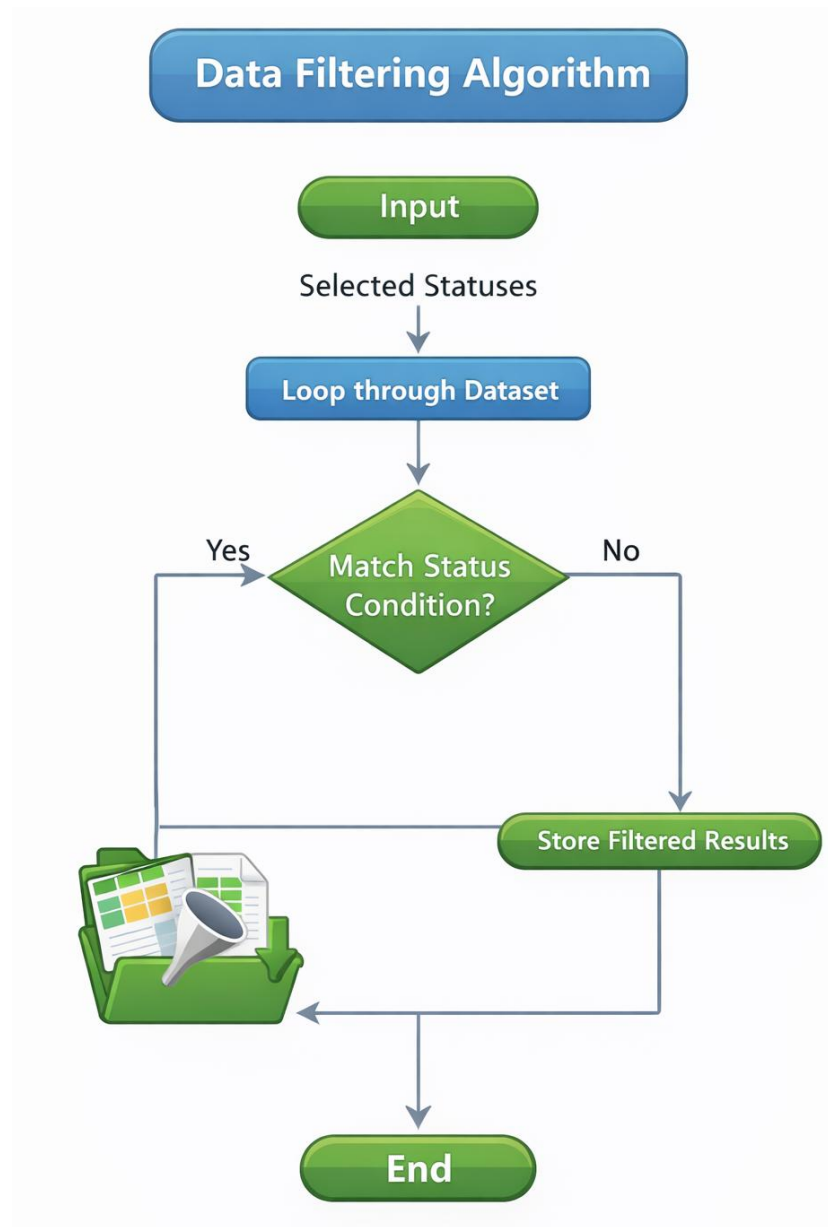


Fig 6.5.5 Data Filtering Algorithm

This algorithm helps users quickly access relevant data without going through the entire dataset. It improves efficiency and supports better data analysis by focusing only on required information.

Advantages

- Saves time by reducing data search effort
- Provides quick access to relevant records
- Improves data analysis efficiency
- Supports better decision-making

6.5.6 Report Generation Algorithm

Process:

- Convert filtered data → JSON
- Convert JSON → Excel sheet
- Generate file buffer
- Send as downloadable file

The Report Generation Algorithm is responsible for creating downloadable reports from filtered or processed data. It converts the selected dataset into JSON format, then transforms it into an Excel file. Finally, the system generates a file buffer and sends it to the user for download.



Fig 6.5.6 Report Generation Algorithm

This algorithm automates the reporting process, eliminating manual effort and ensuring accuracy. It allows users to generate customized reports based on their requirements, which can be used for analysis and sharing.

Advantages

- Automates report creation process
- Ensures accurate and consistent reports
- Supports customized report generation
- Improves productivity and efficiency

CHAPTER – 7

APPENDICES

7.1 ER DIAGRAM



Fig 7.1 ER Diagram

7.2 USE CASE DIAGRAM

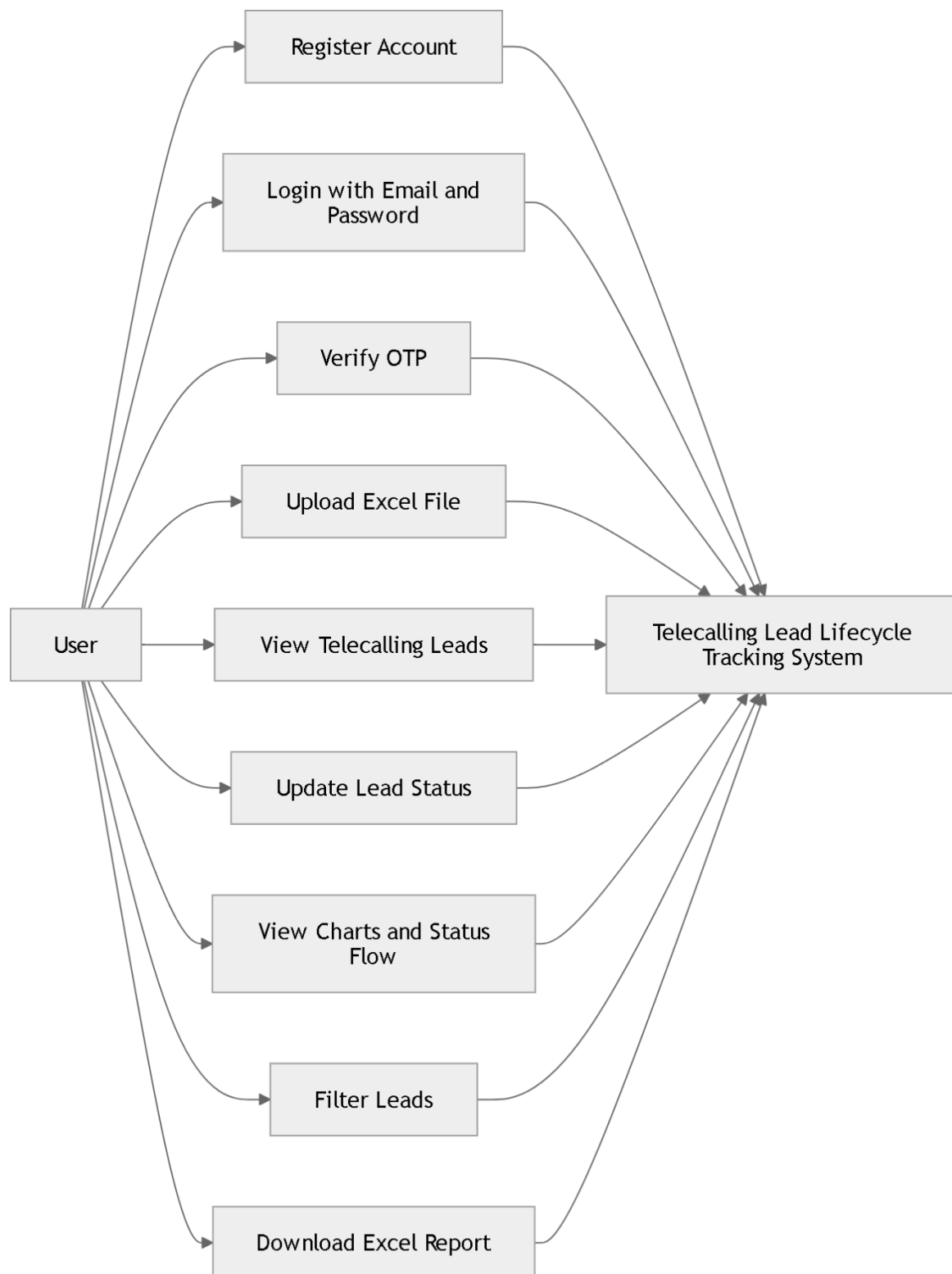


Fig 7.2 Use Case Diagram

7.3 CLASS DIAGRAM

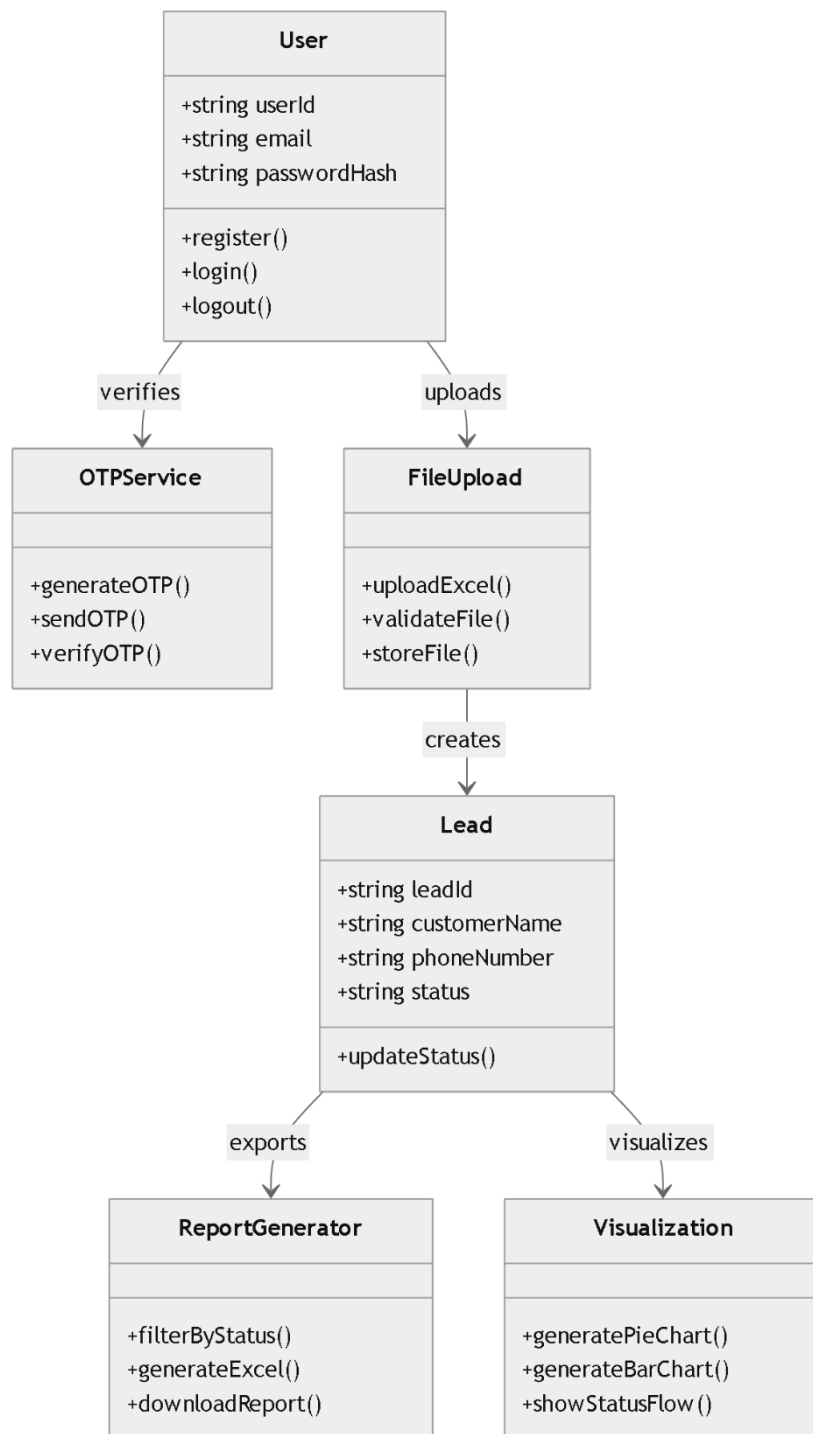
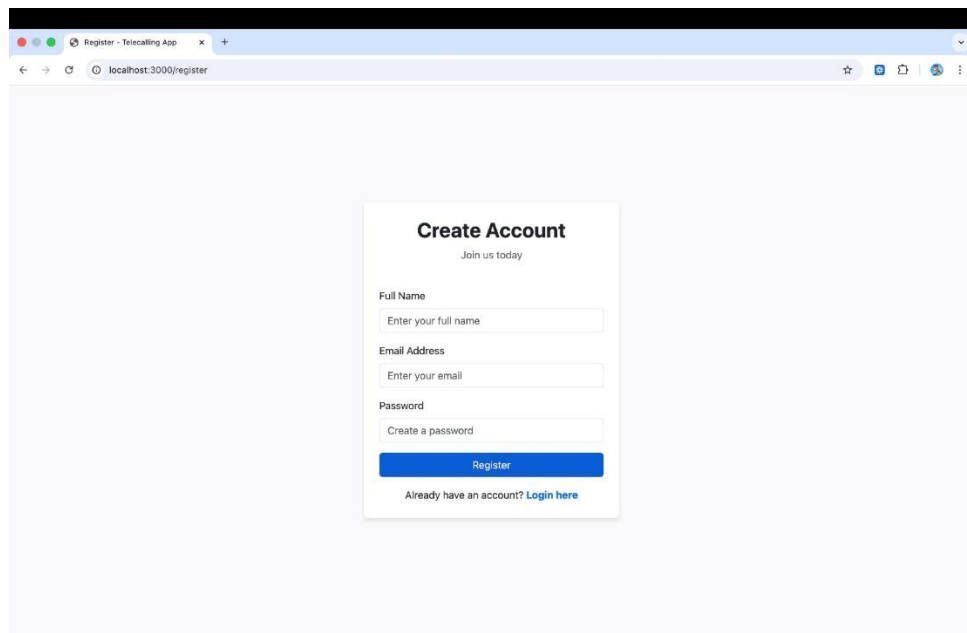


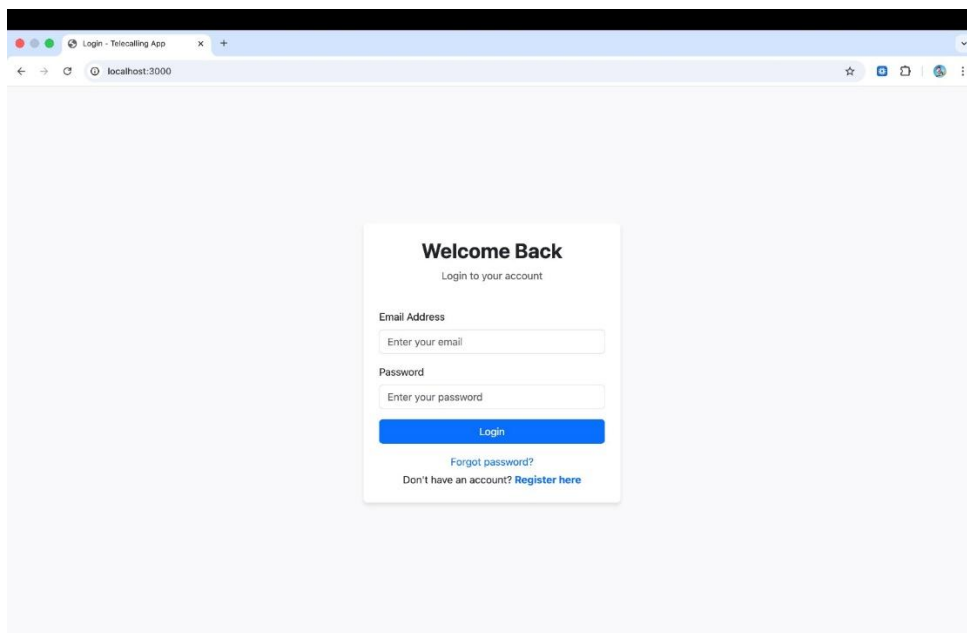
Fig 7.3 Class Diagram

7.4 SCREENSHOTS



A screenshot of a web browser showing the 'Register' page. The browser's address bar displays 'localhost:3000/register'. The page features a central white card with the title 'Create Account' and the subtitle 'Join us today'. Below the title, there are three input fields: 'Full Name' with the placeholder 'Enter your full name', 'Email Address' with the placeholder 'Enter your email', and 'Password' with the placeholder 'Create a password'. A blue 'Register' button is positioned below the password field. At the bottom of the card, there is a link that says 'Already have an account? [Login here](#)'.

Fig 7.4.1 Register Page



A screenshot of a web browser showing the 'Login' page. The browser's address bar displays 'localhost:3000'. The page features a central white card with the title 'Welcome Back' and the subtitle 'Login to your account'. Below the title, there are two input fields: 'Email Address' with the placeholder 'Enter your email' and 'Password' with the placeholder 'Enter your password'. A blue 'Login' button is positioned below the password field. At the bottom of the card, there are two links: 'Forgot password?' and 'Don't have an account? [Register here](#)'.

Fig 7.4.2 Login Page

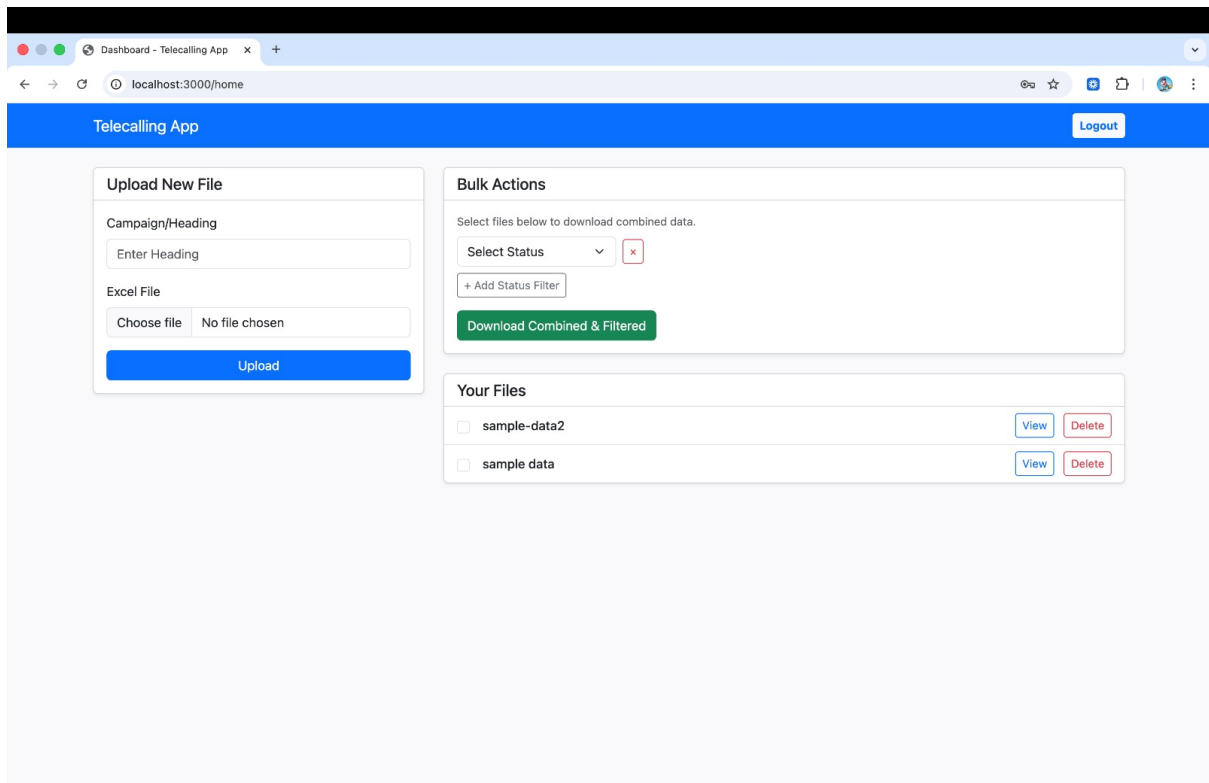


Fig 7.4.3 Dashboard

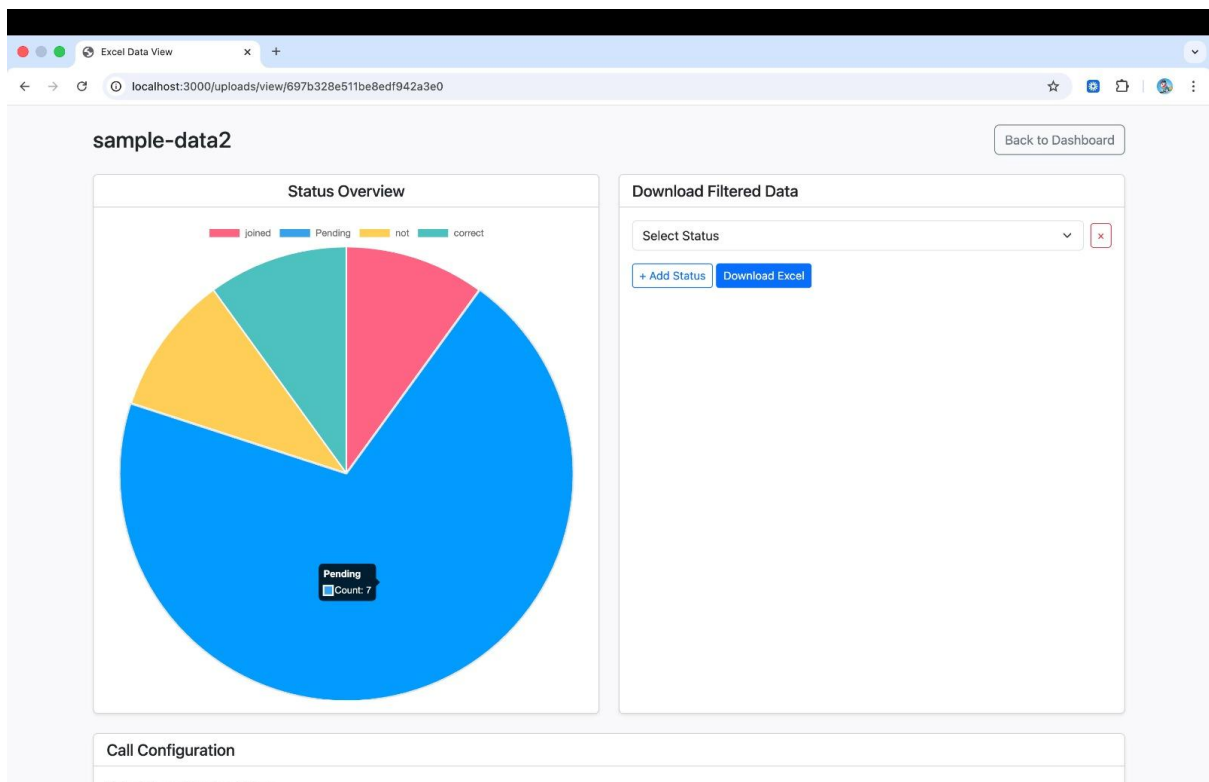


Fig 7.4.4 Chart View for the List

Call Configuration

Select Phone Number Column:

Mobile Number 2

Select the column that contains the phone number to enable the "Call" button.

Data Table

First Name	Last Name	School Name	Mobile Number	Mobile Number 2	Course	status	Action	Call
Lakshmi	Krishnan	Silver Oaks School	9811122233	9011223344	Mechanical Engineering	joined	Update Status	Call
Aditya	Verma	Oakridge International	9822334455	9022334455	Electrical Engineering	Pending	Update Status	Call
Meera	Rao	Baldwin Boys School	9833445566	9033445566	Civil Engineering	Pending	Update Status	Call
Karthik	Chopra	St. Joseph's School	9844556677	9044556677	Electronics	not	Update Status	Call
Sanjana	Desai	Mount Carmel School	9855667788	9055667788	Information Technology	correct	Update Status	Call
Ravi	Bhatia	St. Xavier's High	9866778899	9066778899	Business Administration	Pending	Update Status	Call
Nisha	Gupta	Bethany High	9877889900	9077889900	Psychology	Pending	Update Status	Call
Harish	Shetty	Bishop Cotton	9888990011	9088990011	Fine Arts	Pending	Update Status	Call
Pooja	Naidu	Frank Anthony Public	9899001122	9099001122	Sociology	Pending	Update Status	Call
Suresh	Kulkarni	Presidency School	9900112233	9000112233	Philosophy	Pending	Update Status	Call

Fig 7.4.5 Telecalling List

Telecalling App

Upload New File

Campaign/Heading

Enter Heading

Excel File

Choose file No file chosen

[Upload](#)

Bulk Actions

Select files below to download combined data.

Pending

+ Add Status Filter

[Download Combined & Filtered](#)

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combined_filtered_data (1)
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Fig 7.4.6 Download Multi-Sheet

7.5 SOURCE CODE

Server.js code

```
const express = require("express");

const mongoose = require("mongoose");

const path = require("path");

const session = require("express-session");

const bodyParser = require("body-parser");

require("dotenv").config();

const app= express();

// middlewares first

// parse JSON + form data

app.use(bodyParser.json());

app.use(bodyParser.urlencoded({ extended: true }));

app.use(session({

  secret: "secretkey",

  resave: false,

  saveUninitialized: true

}));
```

```
// 2. static files

app.use(express.static(path.join(__dirname, "public")));

// 3. connect DB

mongoose.connect(process.env.MONGO_URI).then(() => {

  console.log("mongodb connected");

}).catch(error => console.log(error));

// 4. import routes AFTER session middleware

const authRoutes = require("./routes/auth");

app.use("/", authRoutes);

const uploadRoutes = require("./routes/upload");

app.use("/", uploadRoutes);

// 5. start server

app.listen(3000, () => console.log("server running on http://localhost:3000));
```

auth.js code

```
const express = require("express");

const bcrypt = require("bcryptjs");

const User = require("../models/User");

const path = require("path")
```



```

const nodemailer = require("nodemailer")

require("dotenv").config();

const router = express.Router();

router.get("/", (req,res)=>{

    if(req.session.user){

        return res.redirect("/home")

    }

    res.sendFile(path.join(__dirname, "../public/views/login.html"))

})

router.get("/register", (req,res)=>{

    res.sendFile(path.join(__dirname,"../public/views", "register.html"))

})

router.get("/home", (req,res)=>{

    if (req.session.user) {

        res.sendFile(path.join(__dirname, "../public/views", "home.html"));

    } else {

        res.send(

            <script>

```

```

    alert("Please login first");

    window.location.href="/";

</script>

`);

}

})

// API to get logged-in user info

router.get("/user", async (req, res) => {

  if (!req.session.user) {

    return res.status(401).json({ error: "Not logged in" });

  }

  const user = await User.findById(req.session.user).select("name email");

  res.json(user);

});

```

CHAPTER – 8

CONCLUSION & FUTURE ENHANCEMENTS

8.1 CONCLUSION

The Telecalling Lead Lifecycle Tracking System with Status Flow Visualization and Report Download Facility has been successfully designed and implemented to address the challenges faced in traditional telecalling management systems. The system provides a comprehensive solution that simplifies the process of handling telecalling data, improves efficiency, and enhances decision-making capabilities.

The developed system effectively replaces manual and spreadsheet-based approaches by introducing automation and real-time data management. It enables users to upload telecalling data in Excel format, view it in a structured manner, and update lead statuses dynamically. This ensures that all data is maintained accurately and consistently, reducing the chances of human error.

One of the key strengths of the system is its ability to provide visual representation of data through charts and graphical analysis. This feature allows users to quickly understand the distribution of leads across different stages such as Interested, Follow-up, Converted, and Not Interested. Such insights help organizations evaluate performance and improve telecalling strategies.

The system also includes advanced filtering and reporting capabilities, allowing users to extract specific datasets and download them as Excel reports. The ability to combine multiple datasets into a single report further enhances its usefulness for analysis and reporting purposes.

Security has been given high importance in the system through features such as user authentication, password encryption, and session management. These ensure that only authorized users can access sensitive data, maintaining data privacy and integrity.

Overall, the system provides a user-friendly, efficient, and scalable solution for telecalling lead management. It successfully meets the objectives defined at the beginning of the project and demonstrates how modern web technologies can be used to improve business processes.

8.2 FUTURE ENHANCEMENTS

Although the proposed system provides a robust solution for telecalling management, there are several areas where further improvements and enhancements can be made to increase its functionality and performance.

One of the major enhancements can be the integration of cloud-based storage. Currently, the system stores files locally, but moving to cloud platforms can improve scalability, accessibility, and data security. This will allow users to access the system from anywhere without dependency on a local server.

Another important enhancement is the addition of role-based access control. In future versions, the system can be designed to support multiple user roles such as admin, manager, and telecaller, each with different levels of access and permissions. This will improve system security and usability in large organizations.

The system can also be enhanced by integrating real-time calling features using APIs. This would allow telecallers to make calls directly from the system and automatically update call status, reducing manual effort.

Artificial Intelligence and Machine Learning techniques can be introduced to analyze telecalling data and provide predictive insights. For example, the system can suggest the probability of lead conversion or recommend follow-up strategies based on past data.

Another possible improvement is the implementation of advanced dashboards with interactive analytics. These dashboards can provide deeper insights into performance metrics, helping managers make better decisions.

Mobile application support is also a valuable enhancement. Developing a mobile version of the system will enable users to manage telecalling operations on the go, increasing flexibility and productivity.

Additionally, features such as automated reminders for follow-ups, integration with email and SMS services, and multilingual support can further improve the usability of the system.

In conclusion, while the current system meets all essential requirements, future enhancements can make it more powerful, scalable, and suitable for real-world enterprise applications.

CHAPTER – 9

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